

~ Calibration Certificate ~

Model Number: 480B21

Customer: _____

Serial Number: 41481

Description: 3 Channel Signal Conditioner

P.O.: _____

Manufacturer: PCB

Method: Comparison Method (AT103-19)

Calibration Data

Temperature: 75 °F (24 °C)

Humidity: 43%

Channel	Volts	Current (mA)	Gain X1	Gain X10	Gain X100
1	27.3	2.97	1.001	10.011	100.1
2	26.4	2.98	1.001	10.008	100.1
3	26.4	2.97	1.001	10.013	100.1

Condition of Unit

(Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.)

As Found: N/A

As Left: New unit, in tolerance

Notes

1. Calibration is N.I.S.T. traceable through PCB control number QC726.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSS Z540.3 and ISO 17025.
4. Measurement uncertainty (95% confidence level with a coverage factor of 2) for DC Volts and AC Volts/Gain is +/- 0.13% of reading.
5. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.

Technician: Hannah Lynch



Date: 02/17/25

Due Date: _____



CALIBRATION CERT #1862.02



Headquarters: 3425 Walden Avenue, Depew, NY 14043

Calibration Performed at: 10869 Highway 903, Halifax, NC 27839

TEL: 888-684-0013

FAX: 716-685-3886

www.pcb.com



~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 352C22

Serial Number: LW521801

Description: ICP® Accelerometer

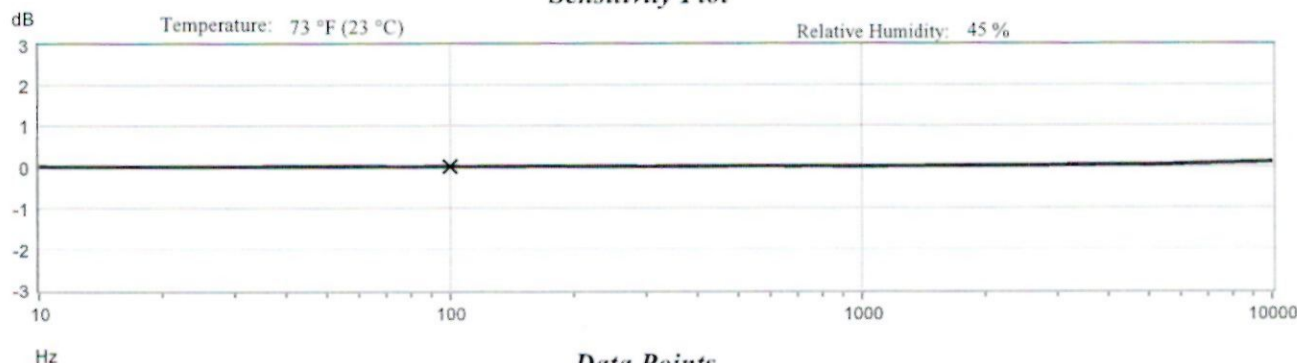
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	10.56 mV/g (1.077 mV/m/s ²)	Output Bias	10.5 VDC
Discharge Time Constant	2.1 seconds	Transverse Sensitivity	2.1 %
		Resonant Frequency	82.5 kHz

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	0.0	300	-0.0	7000	0.7
15	-0.1	500	0.1	10000	1.2
30	-0.0	1000	0.0		
50	0.0	3000	0.3		
REF. FREQ.	0.0	5000	0.4		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical
Acceleration Level (pk): 10.0 g (98.1 m/s²)

The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

- Calibration is NIST Traceable thru Project 684/O-0000035821 and PTB Traceable thru Project 17016.
- This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
- Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
- Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
- Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows:
5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Alexander Stanley

Alexander Stanley



Date: 2/27/2025



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HEADQUARTERS: 3425 WALDEN AVENUE - DEPEW, NY 14043
CALIBRATION PERFORMED AT: 10869 HIGHWAY 903 - HALIFAX, NC 27839
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CAL 48-3823557793.510+0



~ Certificate of Calibration and Compliance ~

Model : 378B02 **Manufacturer :** PCB
Serial : 173314 **Description :** 1/2" Free-Field Microphone and Preamplifier
Microphone Model : 377B02 **Serial :** 364175 **Manufacturer :** PCB
Preamplifier Model : 426E01 **Serial :** 090621 **Manufacturer :** PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PC1e-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0146	CA2115	07/11/2024	07/11/2025
Larson Davis	PRM902	4701	CA1450	07/11/2024	07/11/2025
Larson Davis	PRM916	0129	CA1084	08/16/2024	08/15/2025
Larson Davis	CAL250	4147	LD018	07/08/2024	07/08/2025
Larson Davis	2201	151	CA2073	09/05/2024	09/05/2025
Larson Davis	GPRM902	5337	CA2153	10/25/2024	10/24/2025
Larson Davis	PRM915	132	CA1552	09/11/2024	09/11/2025
Larson Davis	PRA951-4	0241	CA1449	08/17/2024	08/15/2025
Bruel & Kjaer	4192	3259548	CA3533	10/11/2024	10/10/2025
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.20 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik  Date: 03/05/2025



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~ Calibration Report ~

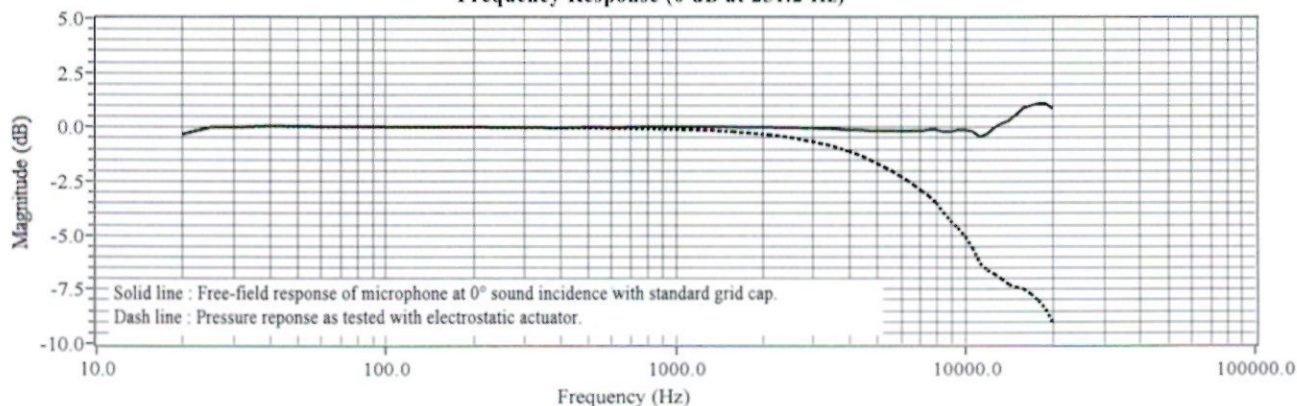
Model : 378B02 **Manufacturer :** PCB
Serial : 173314 **Description :** 1/2" Free-Field Microphone and Preamplifier
Microphone Model : 377B02 **Serial :** 364175 **Manufacturer :** PCB
Preamplifier Model : 426E01 **Serial :** 090621 **Manufacturer :** PCB

Calibration Data

System Sensitivity at 251.2 Hz : 53.67 mV/Pa **Polarization Voltage, External :** 0 V
 -25.41 dB re 1 V/Pa

Temperature: 68 °F (20 °C) **Ambient Pressure:** 976 mbar **Relative Humidity:** 36 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)			
20.00	-0.31	-0.31	1584.90	-0.20	0.01	6683.40	-2.69	-0.17			
25.10	0.02	0.02	1678.80	-0.23	-0.00	7079.50	-2.94	-0.16			
31.60	0.02	0.02	1778.30	-0.25	0.00	7498.90	-3.18	-0.11			
39.80	0.06	0.06	1883.60	-0.28	-0.00	7943.30	-3.49	-0.10			
50.10	0.04	0.04	1995.30	-0.31	-0.00	8414.00	-3.93	-0.20			
63.10	0.03	0.03	2113.50	-0.34	0.00	8912.50	-4.31	-0.20			
79.40	0.02	0.02	2238.70	-0.38	-0.01	9440.60	-4.64	-0.12			
100.00	0.02	0.02	2371.40	-0.42	-0.01	10000.00	-5.07	-0.12			
125.90	0.01	0.01	2511.90	-0.48	-0.02	10592.50	-5.58	-0.18			
158.50	0.01	0.01	2660.70	-0.52	-0.01	11220.20	-6.29	-0.43			
199.50	0.01	0.01	2818.40	-0.60	-0.04	11885.00	-6.62	-0.30			
251.20	0.00	0.00	2985.40	-0.65	-0.03	12589.30	-6.79	-0.02			
316.20	-0.00	0.01	3162.30	-0.74	-0.06	13335.20	-7.01	0.18			
398.10	-0.01	-0.01	3349.70	-0.81	-0.07	14125.40	-7.28	0.31			
501.20	-0.02	0.02	3548.10	-0.89	-0.07	14962.40	-7.40	0.57			
631.00	-0.04	0.00	3758.40	-1.00	-0.10	15848.90	-7.47	0.88			
794.30	-0.06	0.03	3981.10	-1.11	-0.11	16788.00	-7.71	1.01			
1000.00	-0.09	0.03	4217.00	-1.24	-0.13	17782.80	-8.02	1.09			
1059.30	-0.10	0.03	4466.80	-1.37	-0.14	18836.50	-8.42	1.09			
1122.00	-0.10	0.04	4731.50	-1.53	-0.16	19952.60	-9.07	0.86			
1188.50	-0.12	0.03	5011.90	-1.69	-0.16						
1258.90	-0.13	0.03	5308.80	-1.86	-0.16						
1333.50	-0.14	0.04	5623.40	-2.04	-0.16						
1412.50	-0.16	0.03	5956.60	-2.25	-0.18						
1496.20	-0.18	0.02	6309.60	-2.46	-0.17						

Technician: Leonard Lukasik  **Date:** 03/05/2025



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~ Calibration Certificate ~

Model Number: 086E80 Customer: _____
Serial Number: 52563 _____
Description: Impulse Force Hammer _____
Manufacturer: PCB P.O. Number: _____
Method: Impulse (AT303-1)

Calibration Data

Output Bias: 10.5 VDC
Temperature: 73 °F = 23 °C
Relative Humidity: 39 %

Hammer Configuration

Hammer Sensitivity*

Tip	Extender	mV/lb	mV/N
Steel	None	110.9	24.94
Vinyl	Steel	91.71	20.62
Vinyl	None	85.44	19.21

*Above data is valid for all supplied tips.

Condition of Unit

As Found: N/A
As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Notes:

1. Calibration is N.I.S.T. Traceable through Project 684/O-0000069046 and PTB Traceable through Project 17036.
2. This certificate may not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NC SL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with a coverage factor of 2) is +/-3.8%.

Technician: Mary Warren *MW* Date: 3/4/2025



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~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 352A73

Serial Number: LW521679

Description: ICP® Accelerometer

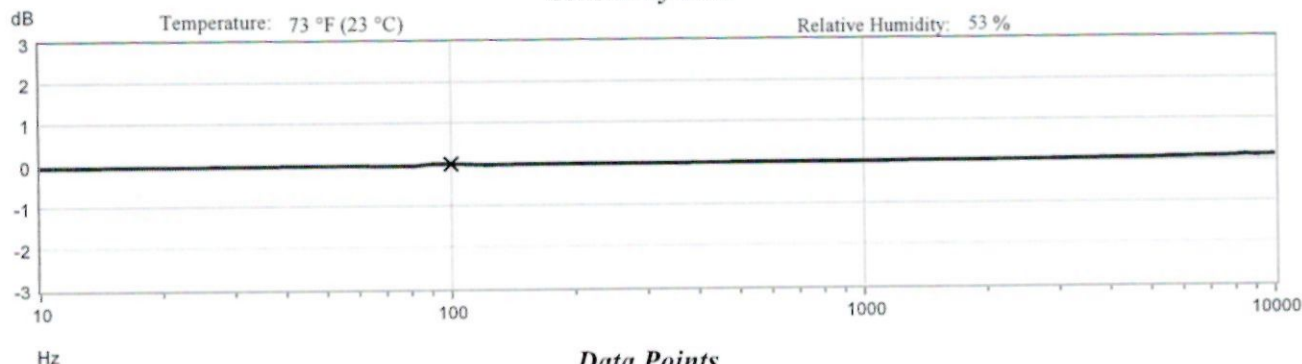
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	5.58 mV/g	Output Bias	9.3 VDC
	(0.569 mV/m/s ²)	Transverse Sensitivity	0.7 %
Discharge Time Constant	0.5 seconds	Resonant Frequency	108.1 kHz

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-0.7	300	-0.2	7000	0.9
15	-0.6	500	-0.0	10000	1.2
30	-0.5	1000	0.0		
50	-0.4	3000	0.3		
REF. FREQ.	0.0	5000	0.6		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical
Acceleration Level (pk): 10.0 g (98.1 m/s²)

¹The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

²The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

1. Calibration is NIST Traceable thru Project 684/O-0000035821 and PTB Traceable thru Project 17016.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows:
5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician:

Jaquan Robinson

Date: 3/10/2025



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~ Certificate of Calibration and Compliance ~

Model : 378C10 **Manufacturer :** PCB
Serial : 175491 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 363727 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 082728 **Manufacturer :** PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	2699	TA468	12/27/2024	12/26/2025
Larson Davis	PRM916	0129	CA1084	08/16/2024	08/15/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	151	CA2073	09/05/2024	09/05/2025
Larson Davis	GPRM902	5337	CA2153	10/25/2024	10/24/2025
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	08/17/2024	08/15/2025
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSS Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik Date: 07/14/2025



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~ Calibration Report ~

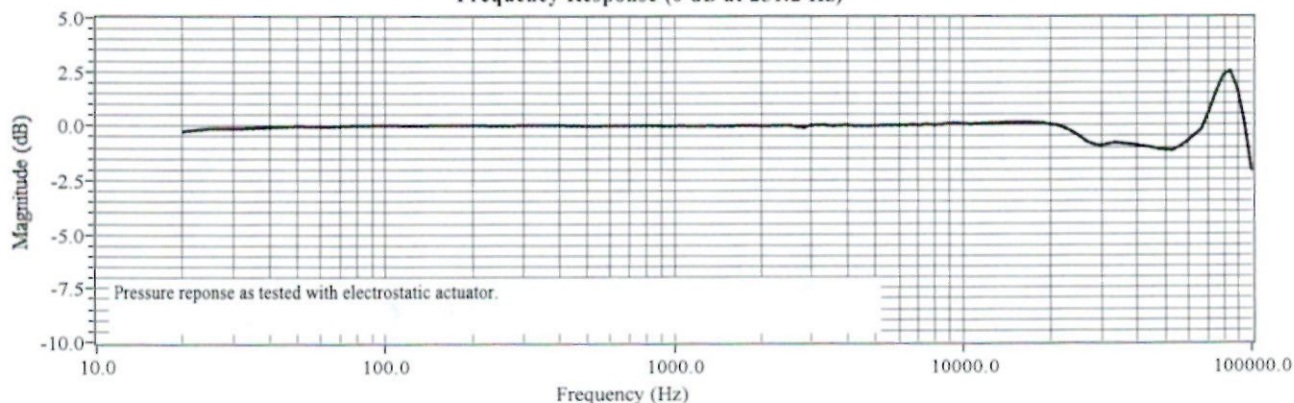
Model : 378C10 **Manufacturer :** PCB
Serial : 175491 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 363727 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 082728 **Manufacturer :** PCB

Calibration Data

System Sensitivity at 251.2 Hz : 0.97 mV/Pa **Polarization Voltage, External :** 0 V
 -60.27 dB re 1 V/Pa

Temperature: 70 °F (21 °C) **Ambient Pressure:** 991 mbar **Relative Humidity:** 49 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.25	1584.90	-0.02	6683.40	0.03	28250.00	-0.88
25.10	-0.12	1678.80	-0.00	7079.50	0.02	29750.00	-0.93
31.60	-0.12	1778.30	-0.01	7498.90	0.05	31500.00	-0.88
39.80	-0.07	1883.60	0.00	7943.30	0.03	33500.00	-0.81
50.10	-0.02	1995.30	-0.02	8414.00	0.05	35500.00	-0.85
63.10	-0.04	2113.50	-0.01	8912.50	0.09	37500.00	-0.88
79.40	0.00	2238.70	-0.01	9440.60	0.10	39750.00	-0.92
100.00	0.01	2371.40	0.00	10000.00	0.09	42250.00	-0.98
125.90	0.00	2511.90	0.01	10592.50	0.06	44750.00	-1.02
158.50	0.01	2660.70	-0.04	11220.20	0.07	47250.00	-1.10
199.50	0.01	2818.40	-0.09	11885.00	0.10	50000.00	-1.13
251.20	0.00	2985.40	0.02	12589.30	0.10	53000.00	-1.15
316.20	0.02	3162.30	0.04	13335.20	0.10	56250.00	-0.98
398.10	0.01	3349.70	0.02	14125.40	0.13	59500.00	-0.76
501.20	-0.02	3548.10	-0.02	14962.40	0.14	63000.00	-0.47
631.00	-0.01	3758.40	0.01	15848.90	0.13	66750.00	-0.20
794.30	-0.01	3981.10	0.03	16788.00	0.13	70750.00	0.60
1000.00	-0.02	4217.00	-0.01	17782.80	0.12	75000.00	1.54
1059.30	-0.01	4466.80	-0.02	18836.50	0.10	79500.00	2.31
1122.00	-0.01	4731.50	-0.01	19952.60	0.04	84250.00	2.52
1188.50	-0.01	5011.90	0.00	21250.00	-0.00	89250.00	1.66
1258.90	-0.02	5308.80	0.01	22500.00	-0.14	94500.00	0.08
1333.50	-0.01	5623.40	0.01	23750.00	-0.28	100000.00	-2.09
1412.50	-0.02	5956.60	0.02	25000.00	-0.47		
1496.20	-0.02	6309.60	0.01	26500.00	-0.72		

Technician: Leonard Lukasik **Date:** 07/14/2025



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~ Certificate of Calibration and Compliance ~

Model : 378C10 Manufacturer : PCB
Serial : 175492 Description : 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 Serial : 363725 Manufacturer : PCB
Preamplifier Model : 426B03 Serial : 082729 Manufacturer : PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	2699	TA468	12/27/2024	12/26/2025
Larson Davis	PRM916	0129	CA1084	08/16/2024	08/15/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	151	CA2073	09/05/2024	09/05/2025
Larson Davis	GPRM902	5337	CA2153	10/25/2024	10/24/2025
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	08/17/2024	08/15/2025
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSS Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik Date: 07/14/2025



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~ Calibration Report ~

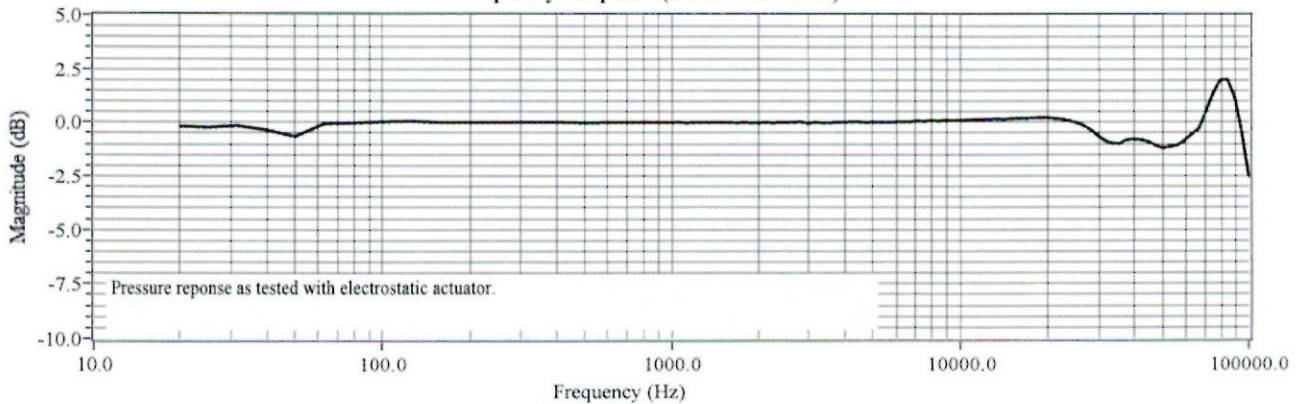
Model : 378C10 **Manufacturer :** PCB
Serial : 175492 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 363725 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 082729 **Manufacturer :** PCB

Calibration Data

System Sensitivity at 251.2 Hz : 1.01 mV/Pa **Polarization Voltage, External :** 0 V
 -59.92 dB re 1 V/Pa

Temperature: 70 °F (21 °C) **Ambient Pressure:** 991 mbar **Relative Humidity:** 49 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.17	1584.90	0.00	6683.40	0.03	28250.00	-0.34
25.10	-0.21	1678.80	-0.01	7079.50	0.07	29750.00	-0.59
31.60	-0.15	1778.30	0.00	7498.90	0.05	31500.00	-0.84
39.80	-0.35	1883.60	-0.01	7943.30	0.09	33500.00	-0.97
50.10	-0.66	1995.30	-0.02	8414.00	0.07	35500.00	-1.00
63.10	-0.06	2113.50	-0.01	8912.50	0.10	37500.00	-0.82
79.40	-0.03	2238.70	-0.02	9440.60	0.10	39750.00	-0.78
100.00	0.03	2371.40	0.00	10000.00	0.10	42250.00	-0.81
125.90	0.05	2511.90	-0.01	10592.50	0.12	44750.00	-0.90
158.50	0.01	2660.70	0.02	11220.20	0.13	47250.00	-1.08
199.50	0.01	2818.40	0.04	11885.00	0.13	50000.00	-1.19
251.20	0.00	2985.40	-0.03	12589.30	0.15	53000.00	-1.14
316.20	0.02	3162.30	-0.01	13335.20	0.16	56250.00	-1.06
398.10	0.01	3349.70	-0.02	14125.40	0.15	59500.00	-0.86
501.20	-0.02	3548.10	-0.03	14962.40	0.18	63000.00	-0.56
631.00	0.01	3758.40	0.01	15848.90	0.19	66750.00	-0.30
794.30	-0.00	3981.10	0.01	16788.00	0.19	70750.00	0.51
1000.00	-0.00	4217.00	0.04	17782.80	0.21	75000.00	1.38
1059.30	0.01	4466.80	0.03	18836.50	0.23	79500.00	1.99
1122.00	-0.01	4731.50	0.00	19952.60	0.22	84250.00	2.01
1188.50	0.01	5011.90	0.01	21250.00	0.17	89250.00	1.15
1258.90	-0.01	5308.80	0.02	22500.00	0.15	94500.00	-0.47
1333.50	0.01	5623.40	0.02	23750.00	0.09	100000.00	-2.55
1412.50	-0.01	5956.60	0.03	25000.00	0.00		
1496.20	0.01	6309.60	0.03	26500.00	-0.12		

Technician: Leonard Lukasik **Date:** 07/14/2025



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~ Calibration Certificate ~

Model Number: 086C01 Customer: _____
Serial Number: 52635 _____
Description: Impulse Force Hammer _____
Manufacturer: PCB P.O. Number: _____
Method: Impulse (AT303-1)

Calibration Data

Output Bias: 10.2 VDC

Temperature: 73 °F = 23 °C

Relative Humidity: 43 %

Hammer Configuration

Hammer Sensitivity*

Tip	Extender	mV/lb	mV/N
Plastic/Vinyl with tuning mass	None	51.65	11.61
Plastic/Vinyl with tuning mass	Aluminum	53.29	11.98
Plastic/Vinyl	None	47.46	10.67

*Above data is valid for all supplied tips.

Condition of Unit

As Found: N/A
As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Notes:

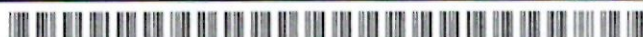
1. Calibration is N.I.S.T. Traceable through Project 684/O-0000069046 and PTB Traceable through Project 17036.
2. This certificate may not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCISL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with a coverage factor of 2) is +/-3.8%.

Technician: Austin Crawley AC Date: 2/26/2025



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~ Certificate of Calibration and Compliance ~

Model : 378C10 **Manufacturer :** PCB
Serial : 178033 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 370168 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 083097 **Manufacturer :** PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PC1e-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	5352	CA1247	12/27/2024	12/26/2025
Larson Davis	PRM916	0144	CA2001	12/27/2024	12/26/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	149	CA2072	05/27/2025	05/27/2026
Larson Davis	GPRM902	4162	CA1088	03/12/2025	03/12/2026
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	09/02/2025	09/02/2026
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

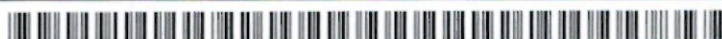
1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSS Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik  Date: 12/16/2025




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~ Calibration Report ~

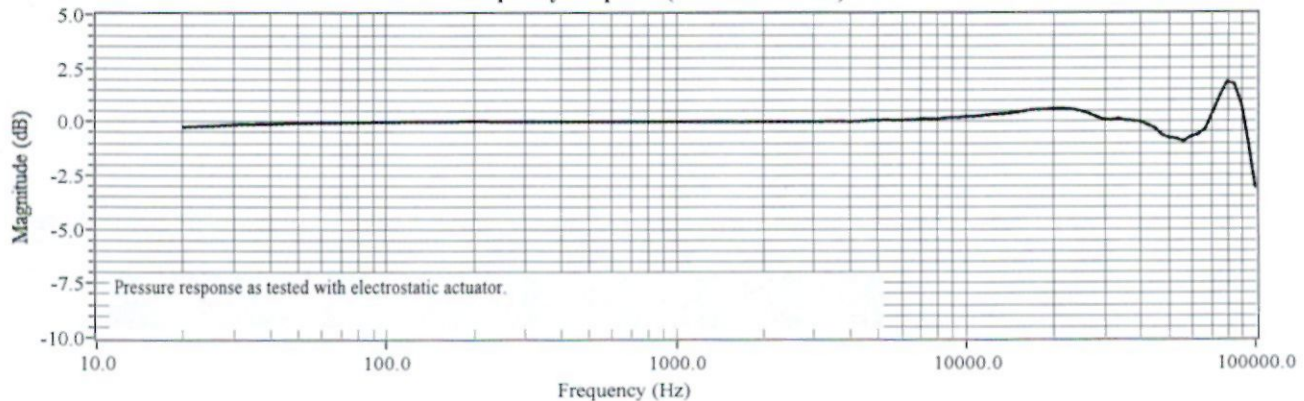
Model : 378C10 **Manufacturer :** PCB
Serial : 178033 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 370168 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 083097 **Manufacturer :** PCB

Calibration Data

System Sensitivity at 251.2 Hz : 0.98 mV/Pa **Polarization Voltage, External :** 0 V
 -60.17 dB re 1 V/Pa

Temperature: 68 °F (20 °C) **Ambient Pressure:** 994 mbar **Relative Humidity:** 30 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.24	1584.90	-0.00	6683.40	0.09	28250.00	0.20
25.10	-0.20	1678.80	-0.02	7079.50	0.13	29750.00	0.10
31.60	-0.12	1778.30	-0.00	7498.90	0.11	31500.00	0.08
39.80	-0.11	1883.60	-0.00	7943.30	0.12	33500.00	0.14
50.10	-0.06	1995.30	0.01	8414.00	0.16	35500.00	0.06
63.10	-0.05	2113.50	-0.00	8912.50	0.18	37500.00	0.04
79.40	-0.03	2238.70	-0.00	9440.60	0.20	39750.00	-0.01
100.00	-0.02	2371.40	-0.00	10000.00	0.23	42250.00	-0.17
125.90	-0.00	2511.90	0.00	10592.50	0.24	44750.00	-0.31
158.50	-0.00	2660.70	0.00	11220.20	0.28	47250.00	-0.62
199.50	0.01	2818.40	0.00	11885.00	0.32	50000.00	-0.75
251.20	0.00	2985.40	0.00	12589.30	0.34	53000.00	-0.79
316.20	0.01	3162.30	-0.01	13335.20	0.35	56250.00	-0.92
398.10	-0.00	3349.70	0.01	14125.40	0.39	59500.00	-0.70
501.20	0.01	3548.10	0.01	14962.40	0.43	63000.00	-0.60
631.00	-0.00	3758.40	0.02	15848.90	0.49	66750.00	-0.36
794.30	0.01	3981.10	0.00	16788.00	0.54	70750.00	0.41
1000.00	-0.00	4217.00	0.01	17782.80	0.56	75000.00	1.21
1059.30	0.00	4466.80	0.04	18836.50	0.58	79500.00	1.83
1122.00	0.01	4731.50	0.06	19952.60	0.61	84250.00	1.74
1188.50	-0.01	5011.90	0.05	21250.00	0.60	89250.00	0.79
1258.90	-0.01	5308.80	0.07	22500.00	0.59	94500.00	-1.02
1333.50	-0.01	5623.40	0.06	23750.00	0.54	100000.00	-3.11
1412.50	-0.00	5956.60	0.05	25000.00	0.48		
1496.20	-0.00	6309.60	0.07	26500.00	0.37		

Technician: Leonard Lukasik **Date:** 12/16/2025



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~ Certificate of Calibration and Compliance ~

Model : 378C10 **Manufacturer :** PCB
Serial : 178034 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 369923 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 083112 **Manufacturer :** PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PC1e-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	5352	CA1247	12/27/2024	12/26/2025
Larson Davis	PRM916	0144	CA2001	12/27/2024	12/26/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	149	CA2072	05/27/2025	05/27/2026
Larson Davis	GPRM902	4162	CA1088	03/12/2025	03/12/2026
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	09/02/2025	09/02/2026
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik  Date: 12/16/2025




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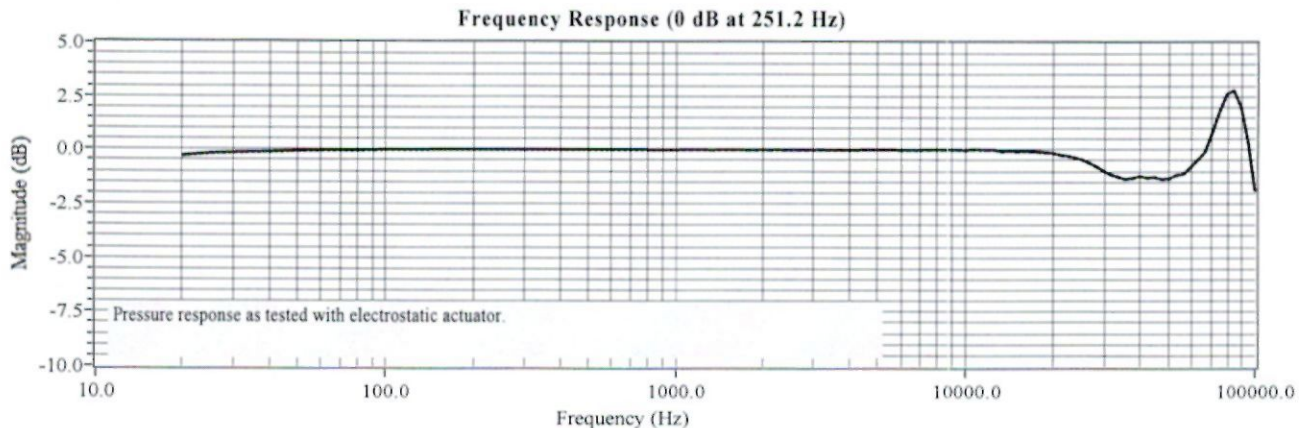
~ Calibration Report ~

Model : 378C10 **Manufacturer :** PCB
Serial : 178034 **Description :** 1/4" Pressure Microphone and Preamplifier
Microphone Model : 377C10 **Serial :** 369923 **Manufacturer :** PCB
Preamplifier Model : 426B03 **Serial :** 083112 **Manufacturer :** PCB

Calibration Data

System Sensitivity at 251.2 Hz : 0.97 mV/Pa **Polarization Voltage, External :** 0 V
 -60.26 dB re 1 V/Pa

Temperature: 68 °F (20 °C) **Ambient Pressure:** 994 mbar **Relative Humidity:** 30 %



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.30	1584.90	-0.01	6683.40	-0.02	28250.00	-0.76
25.10	-0.19	1678.80	0.00	7079.50	-0.01	29750.00	-0.95
31.60	-0.15	1778.30	-0.01	7498.90	-0.01	31500.00	-1.13
39.80	-0.11	1883.60	-0.02	7943.30	-0.00	33500.00	-1.26
50.10	-0.07	1995.30	0.00	8414.00	-0.00	35500.00	-1.36
63.10	-0.04	2113.50	-0.01	8912.50	0.00	37500.00	-1.31
79.40	-0.04	2238.70	-0.01	9440.60	-0.01	39750.00	-1.22
100.00	-0.02	2371.40	-0.02	10000.00	-0.02	42250.00	-1.30
125.90	-0.01	2511.90	-0.01	10592.50	0.01	44750.00	-1.27
158.50	-0.00	2660.70	-0.01	11220.20	0.00	47250.00	-1.36
199.50	-0.00	2818.40	-0.02	11885.00	-0.00	50000.00	-1.35
251.20	0.00	2985.40	-0.01	12589.30	-0.01	53000.00	-1.18
316.20	0.00	3162.30	-0.01	13335.20	-0.08	56250.00	-1.13
398.10	0.00	3349.70	-0.02	14125.40	-0.04	59500.00	-0.81
501.20	-0.00	3548.10	-0.02	14962.40	-0.07	63000.00	-0.47
631.00	0.00	3758.40	-0.02	15848.90	-0.06	66750.00	-0.11
794.30	-0.00	3981.10	-0.02	16788.00	-0.07	70750.00	0.80
1000.00	-0.01	4217.00	-0.02	17782.80	-0.09	75000.00	1.81
1059.30	0.00	4466.80	-0.00	18836.50	-0.11	79500.00	2.60
1122.00	0.00	4731.50	-0.01	19952.60	-0.15	84250.00	2.78
1188.50	-0.00	5011.90	0.00	21250.00	-0.23	89250.00	1.98
1258.90	-0.01	5308.80	-0.00	22500.00	-0.28	94500.00	0.34
1333.50	-0.01	5623.40	0.00	23750.00	-0.36	100000.00	-1.91
1412.50	-0.00	5956.60	-0.02	25000.00	-0.44		
1496.20	-0.00	6309.60	-0.02	26500.00	-0.58		

Technician: Leonard Lukasik **Date:** 12/16/2025



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~ Certificate of Calibration and Compliance ~

Model : 378B02 **Manufacturer :** PCB
Serial : 179741 **Description :** 1/2" Free-Field Microphone and Preamplifier
Microphone Model : 377B02 **Serial :** 372797 **Manufacturer :** PCB
Preamplifier Model : 426E01 **Serial :** 096215 **Manufacturer :** PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	04/17/2026	10/15/2027
Larson Davis	PRM915	0150	CA2116	01/26/2026	01/26/2027
Larson Davis	PRM902	5300	CA1627	11/04/2025	11/04/2026
Larson Davis	PRM916	140	CA2012	11/03/2025	11/03/2026
Larson Davis	CAL250	5026	CA1278	09/04/2025	09/04/2026
Larson Davis	2201	150	CA2228	12/02/2025	12/02/2026
Larson Davis	GPRM902	4923	CA2237	12/01/2025	12/01/2026
Larson Davis	PRM915	147	CA2179	12/02/2025	12/02/2026
Larson Davis	PRA951-4	0241	CA1449	09/02/2025	09/02/2026
Brueel & Kjaer	4192	3259548	CA3533	01/29/2026	01/29/2027
Newport	iTHX-SD/N	1080002	CA1511	03/31/2026	03/31/2027
		N/A	N/A	N/A	N/A

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.20 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik Date: 05/08/2026



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~ Calibration Report ~

Model : 378B02 Manufacturer : PCB
 Serial : 179741 Description : 1/2" Free-Field Microphone and Preamplifier
 Microphone Model : 377B02 Serial : 372797 Manufacturer : PCB
 Preamplifier Model : 426E01 Serial : 096215 Manufacturer : PCB

Calibration Data

System Sensitivity at 251.2 Hz : 49.15 mV/Pa
 -26.17 dB re 1 V/Pa

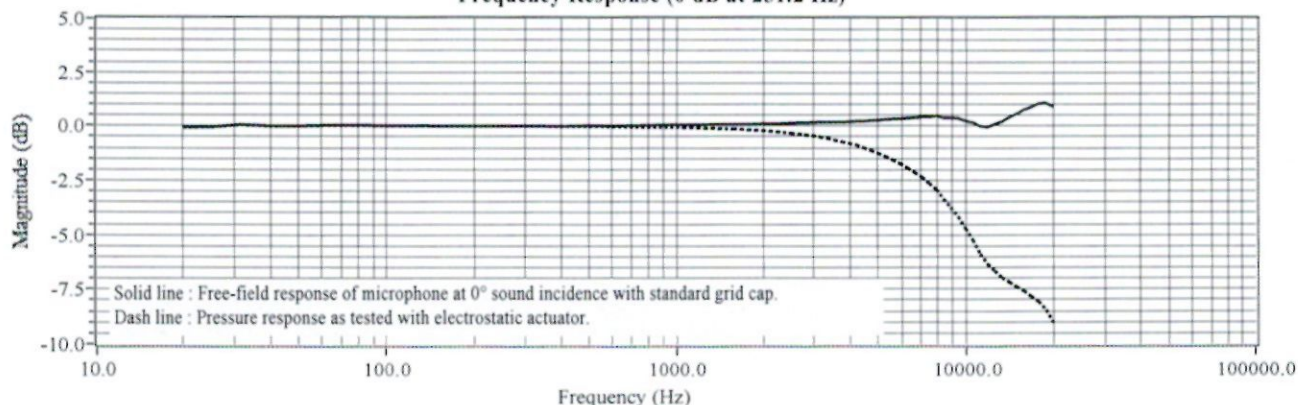
Polarization Voltage, External : 0 V

Temperature: 68 °F (20 °C)

Ambient Pressure: 985 mbar

Relative Humidity: 36 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)
20.00	-0.07	-0.07	1584.90	-0.14	0.07	6683.40	-2.14	0.38
25.10	-0.04	-0.04	1678.80	-0.15	0.08	7079.50	-2.36	0.42
31.60	0.06	0.06	1778.30	-0.18	0.07	7498.90	-2.66	0.41
39.80	0.00	0.00	1883.60	-0.19	0.09	7943.30	-2.93	0.46
50.10	-0.00	-0.00	1995.30	-0.21	0.10	8414.00	-3.36	0.37
63.10	0.02	0.02	2113.50	-0.24	0.10	8912.50	-3.74	0.37
79.40	0.03	0.03	2238.70	-0.27	0.10	9440.60	-4.17	0.35
100.00	0.01	0.01	2371.40	-0.30	0.11	10000.00	-4.72	0.23
125.90	0.01	0.01	2511.90	-0.33	0.13	10592.50	-5.24	0.16
158.50	-0.00	-0.00	2660.70	-0.36	0.15	11220.20	-5.88	-0.02
199.50	0.00	0.00	2818.40	-0.41	0.15	11885.00	-6.37	-0.05
251.20	0.00	0.00	2985.40	-0.45	0.17	12589.30	-6.67	0.10
316.20	-0.01	0.00	3162.30	-0.50	0.18	13335.20	-6.99	0.20
398.10	-0.02	-0.02	3349.70	-0.56	0.18	14125.40	-7.19	0.40
501.20	-0.02	0.02	3548.10	-0.65	0.17	14962.40	-7.40	0.57
631.00	-0.03	0.01	3758.40	-0.72	0.18	15848.90	-7.59	0.76
794.30	-0.05	0.04	3981.10	-0.81	0.19	16788.00	-7.84	0.88
1000.00	-0.07	0.05	4217.00	-0.88	0.23	17782.80	-8.08	1.03
1059.30	-0.08	0.05	4466.80	-1.01	0.22	18836.50	-8.47	1.04
1122.00	-0.08	0.06	4731.50	-1.12	0.25	19952.60	-9.02	0.91
1188.50	-0.09	0.06	5011.90	-1.26	0.27			
1258.90	-0.10	0.06	5308.80	-1.40	0.30			
1333.50	-0.10	0.08	5623.40	-1.56	0.32			
1412.50	-0.12	0.07	5956.60	-1.74	0.33			
1496.20	-0.13	0.07	6309.60	-1.94	0.35			

Technician: Leonard Lukasik

Date: 05/08/2026



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~ Calibration Certificate ~

Model Number: 086E80

Customer: _____

Serial Number: 54385

Description: Impulse Force Hammer

P.O. Number: _____

Manufacturer: PCB

Method: Impulse (AT303-1)

Calibration Data

Output Bias: 10.2 VDC

Temperature: 71 °F = 22 °C

Relative Humidity: 41 %

Hammer Configuration

Hammer Sensitivity*

Tip	Extender	mV/lb	mV/N
Steel	None	99.96	22.47
Vinyl	Steel	84.62	19.03
Vinyl	None	80.87	18.18

*Above data is valid for all supplied tips.

Condition of Unit

As Found: N/A

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Notes:

1. Calibration is N.I.S.T. Traceable through Project 684/O-0000069046 and PTB Traceable through Project 17036.
2. This certificate may not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSS Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with a coverage factor of 2) is +/-3.8%.

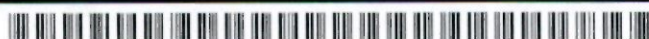
Technician: Richard Gardner RG

Date: 5/11/2026



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~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 356A03

Serial Number: LW458895 (x axis)

Description: ICP® Triaxial Accelerometer

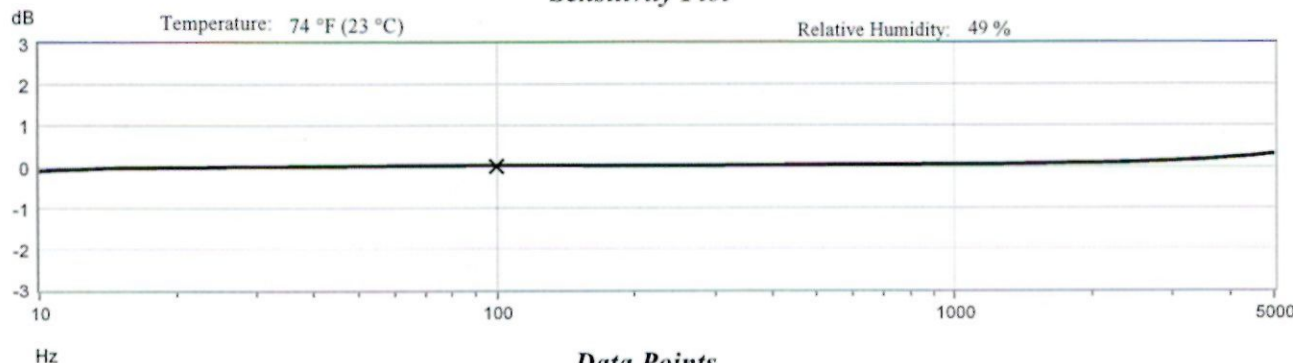
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz 10.34 mV/g
(1.055 mV/m/s²) Output Bias 11.7 VDC
Discharge Time Constant 0.40 seconds Transverse Sensitivity 2.0 %

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-1.2	300	0.2
15	-0.4	500	0.4
30	-0.1	1000	0.6
50	0.1	3000	1.5
REF. FREQ.	0.0	5000	3.6

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Inverted Vertical
Acceleration Level (pk): 10.0 g (98.1 m/s²)

¹The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².
²The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

- Calibration is NIST Traceable thru Project 684/O-0000069046 and PTB Traceable thru Project 17036.
- This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
- Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
- Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
- Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows:
5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

SLE
6124

Technician: _____

Simeon Eason

Date: 5/21/2026



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CAL 48-3862250197.979+0



~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 356A03

Serial Number: LW458895 (y axis)

Description: ICP® Triaxial Accelerometer

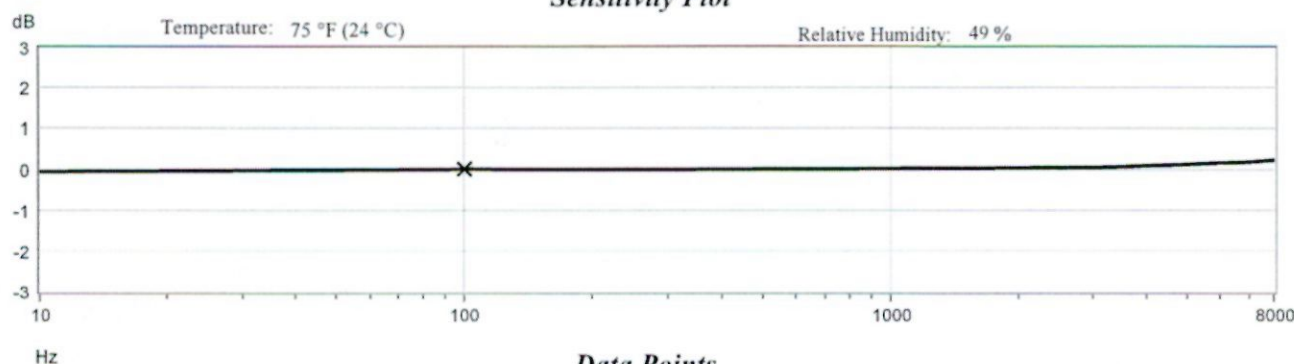
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz 10.25 mV/g
(1.045 mV/m/s²)
Output Bias 11.4 VDC
Transverse Sensitivity 0.4 %
Discharge Time Constant 0.37 seconds

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-0.8	300	-0.2	7000	2.0
15	-0.6	500	0.0	8000	2.5
30	-0.4	1000	0.1		
50	-0.3	3000	0.5		
REF. FREQ.	0.0	5000	1.3		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical

Acceleration Level (pk): 10.0 g (98.1 m/s²)

The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

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3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows:
5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Simeon Eason



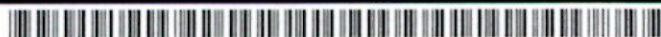
Date: 5/21/2026



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CAL48-3862248076.178+0



~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 356A03

Serial Number: LW458895 (z axis)

Description: ICP® Triaxial Accelerometer

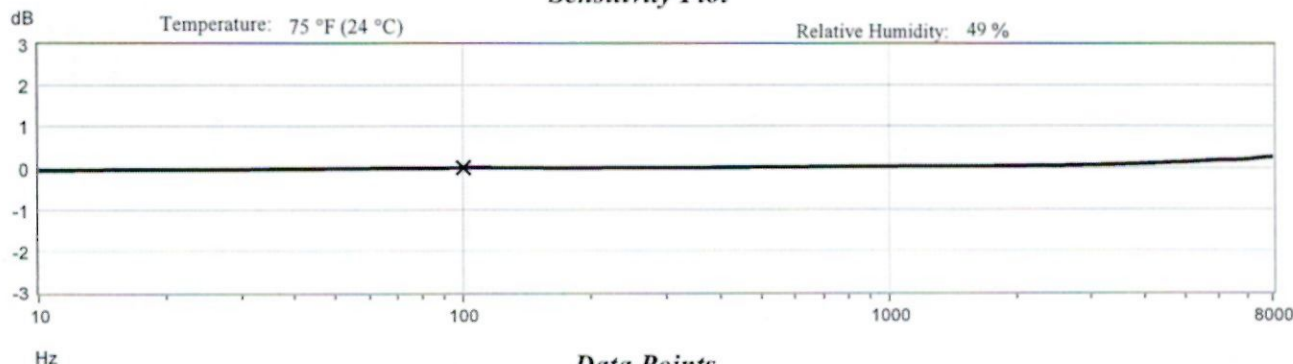
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz 10.94 mV/g Output Bias 11.5 VDC
(1.116 mV/m/s²) Transverse Sensitivity 1.1 %
Discharge Time Constant 0.41 seconds

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-0.7	300	-0.1	7000	2.2
15	-0.6	500	0.1	8000	2.9
30	-0.5	1000	0.2		
50	-0.3	3000	0.7		
REF. FREQ.	0.0	5000	1.5		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical
Acceleration Level (pk): 10.0 g (98.1 m/s²)

¹The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

²The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

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5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows:
5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Simeon Eason

Date: 5/21/2026



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